Hoang Phan

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Education

New York University — New York, USA
Ph.D., Data Science, Advisor: Qi Lei — *Sep. 2023 - Present*

Hanoi University of Science and Technology — Hanoi, Vietnam
B.Sc., Computer Science, Honors program — *Aug. 2018 - Aug. 2022*

Experience

Meta — Bellevue, USA
Research Scientist Intern @ Meta Superintelligence Labs — *June. 2026 - Dec. 2026*
- Developed user simulators and post-training methods for agents using human interactions

Research Scientist Intern @ Meta Superintelligence Labs — *May. 2025 - Oct. 2025*
- Worked on VLMs reasoning and factuality. Developed a post-training method to incorporate multiple rewards

Amazon — Palo Alto, USA
Applied Scientist Intern @ Rufus Team — *May. 2024 - Aug. 2024*
- Proposed a general framework for the data mixture selection that achieved SOTA performance, speeding up the pretraining of LLMs by 35%

Yaraku Inc — Tokyo, Japan *(Remote)*
NLP Engineer — *Feb. 2023 - Aug. 2023*
- Implemented continual learning methods for machine translation to fix reported wrong translations periodically
- Improved the performance of current phrase extraction and sentence alignment systems

VinAI Research (now **Qualcomm AI)** — Hanoi, Vietnam
Research Resident — *Aug. 2021 - Feb. 2023*
- Main research topics: *Robust machine learning, Transfer learning, Multi-task learning, Self-supervised learning*
- Participated in a Smart City project that aimed to build a kidnapping and unshielded truck detection system

Data Science Laboratory — SoICT-HUST
Undergraduate Research Student — *Sep. 2019 - Aug. 2021*
- Main research topics: *Probabilistic inference, Continual learning*
- Developed a Khmer processing toolkit (> 350k downloads). Adopted by Meta Research projects (NLLB, LASER, stopes), NVIDIA Plan2Align, FineWeb 2, Hugging Face datatrove, NAACL Workshop, Wordless and 70+ others

Cinnamon Inc — Hanoi, Vietnam
AI Research Engineer — *Jan. 2020 - June. 2021*
- Worked on information extraction of business documents and meta-learning, question-answering system research
- Developed an internal-use chatbot, integrated with Slack, in support of the Human Resources department

Conference Publications

- **Hoang Phan**, Minh Pham, Chau Pham, Chinmay Hegde, Trung Le, Qi Lei. "Rationale-Guided Policy Optimization: Learning to Reason with Adaptive Rationale Scaffolding". In *Advances in Neural Information Processing Systems (NeurIPS)*, 2026. [pdf]
- **Hoang Phan**, Xianjun Yang, Kevin Yao, Jack Zhang, Shengjie Bi, Xiaocheng Tang, Madian Khabsa, Lijuan Liu, Deren Lei. "Beyond Reasoning Gains: Mitigating General Capabilities Forgetting in Large Reasoning Model". *Findings of the Annual Meeting of the Association for Computational Linguistics (ACL)*, 2026. [pdf]
- Lam Tran, **Hoang Phan**, Christopher Kanan, Trung Le. "RoNE: Robust Neurons Enable Internal Defenses Against Multimodal Jailbreak". *Proceedings of the Annual Meeting of the Association for Computational Linguistics (ACL)*, 2026. [pdf]
- Quyen Tran, Hai Nguyen, Quan Dao, **Hoang Phan**, Linh Ngo, Khoat Than, Dinh Phung, Dimitris Metaxas, Trung Le. "An Optimal Transport-driven Approach for Cultivating Latent Space in Online Incremental Learning". In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026. [pdf]

- Shikai Qiu, Zixi Chen, **Hoang Phan**, Qi Lei, Andrew Gordon Wilson. "Hyperparameter Transfer Enables Consistent Gains of Matrix-Preconditioned Optimizers Across Scales". In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025. [pdf]
- **Hoang Phan**, Victor Li, Qi Lei. "Think Twice, Generate Once: LLMs Safeguarding by Progressive Self-Reflection". In *Findings of the Association for Computational Linguistics: EMNLP*, 2025. [pdf]
- **Hoang Phan***, Lam Tran*, Quyen Tran, Ngoc Tran, Tuan Truong, Qi Lei, Nhat Ho, Dinh Phung, Trung Le. "Beyond Losses Reweighting: Empowering Multi-Task Learning via the Generalization Perspective". In *International Conference on Computer Vision (ICCV)*, **Highlight, Top 2.38%**, 2025. [pdf]
- Long Vuong, **Hoang Phan**, Vy Vo, Anh Bui, Thanh-Toan Do, Trung Le, Dinh Phung. "Preserving Clusters in Prompt Learning for Unsupervised Domain Adaptation". In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025. [pdf]
- Yijun Dong*, **Hoang Phan***, Xiang Pan* and Qi Lei. "Sketchy Moment Matching: Toward Fast and Provable Data Selection for Finetuning". In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. [pdf]
- **Hoang Phan***, Lam Tran*, Quyen Tran* and Trung Le. "Enhancing Domain Adaptation through Prompt Gradient Alignment". In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. [pdf]
- Hao Phung, Quan Dao, Trung Dao, **Hoang Phan**, Dimitris N. Metaxas, Anh Tran. "DiMSUM: Diffusion Mamba - A Scalable and Unified Spatial-Frequency Method For Image Generation." In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. [pdf]
- **Hoang Phan**, Andrew Gordon Wilson, Qi Lei. "Controllable Prompt Tuning for Balancing Group Distributional Robustness". In *International Conference on Machine Learning (ICML)*, 2024. [pdf]
- Anh Nguyen, Long Vuong, **Hoang Phan**, Toan Do, Dinh Phung, Trung Le. "Flat Seeking Bayesian Neural Networks". *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. [pdf]
- **Hoang Phan**, Trung Le, Trung Phung, Anh Bui, Nhat Ho, Dinh Phung. "Global-Local Regularization Via Distributional Robustness". In *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2023. [pdf]
- **Hoang Phan**, Ngoc Tran, Trung Le, Toan Tran, Nhat Ho, Dinh Phung. "Stochastic Multiple Target Sampling Gradient Descent". In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022. [pdf]
- **Hoang Phan***, Anh Phan*, Son Nguyen*, Linh Ngo Van, Khoat Than. "Reducing Catastrophic Forgetting in Neural Networks via Gaussian Mixture Approximation". In *Advances in Knowledge Discovery and Data Mining, PAKDD. Lecture Notes in Computer Science. Springer*, 2022. [pdf]

Workshop & Preprints

- **Hoang Phan**, Dat Huynh, Andrey Zhmoginov, Qi Zeng, Wancen Mu, Yue Cao, Shengjie Bi, Yun He, Changdae Oh, Deren Lei. "MIMESIS: Learning User Simulators as Training Environments for Interactive Agents". Under review. [project page] **[AI Newsletter's Top AI Papers of the Week]**
- Changdae Oh, Qi Zeng, Qi Qi, Andrey Zhmoginov, Deren Lei, Yun He, **Hoang Phan**, Hangoo Kang, Azalia Mirhoseini, Sharon Li. "Sharpening Tax in Post-Training ". Under review. [project page] **[AI Newsletter's Top AI Papers of the Week]**
- **Hoang Phan** et al. "Revisiting Data Mixing Through the Lens of Multi-Objective Optimization". Forever preprint. [pdf]
- **Hoang Phan**, Yijun Dong, Andrew Gordon Wilson, Qi Lei. "Balanced Locality-Sensitive Hashing Sampling for Lightweight Online Data Selection". In *Workshop on Optimization for Machine Learning* at NeurIPS, 2025. [pdf]
- Chau Pham, **Hoang Phan**, David Doermann, Yunjie Tian. "Personalized Large Vision-Language Models". In *Workshop on What is Next in Multimodal Foundation Models?* at CVPR, 2025. [pdf]
- **Hoang Phan**, Sungmin Cha, Lam Tran, Qi Lei. "Toward a Holistic Approach to Continual Model Merging". *Workshop on Continual Learning in Computer Vision* at ICCV, 2025. [pdf]
- Yijun Dong*, Xiang Pan*, **Hoang Phan*** , Qi Lei. "Randomly Pivoted V-optimal Design: Fast Data Selection under Low Intrinsic Dimension". In *Machine Learning and Compression Workshop* at NeurIPS, 2024. [pdf]
- **Hoang Phan**, Long Nguyen, Long Nguyen, Khanh Doan. "Matching The Statements: A Simple and Accurate Model for Key Point Analysis". In *Proceedings of the 8th Workshop on Argument Mining* at EMNLP, 2021. [pdf]
- Tam Nguyen, Quang Pham, Linh Doan, Hoang Trinh, Anh Nguyen, **Hoang Phan**. "Contrastive Learning for Natural Language-Based Vehicle Retrieval". In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops* at CVPR, 2021. [pdf]

(∗) denotes equal contribution

Achievements

ArgMining: Key Point Analysis Shared Task, EMNLP 2021
Presented the 4th solution in the workshop, hosted by IBM Research AI team

NVIDIA AI City Challenge: Natural Language-Based Vehicle Retrieval, CVPR 2021
Placed 4th with a Siamese model capable of extracting visual concepts from natural language

Multimodal Emotion Recognition on Comics scenes, ICDAR 2021
Won the first prize with a late fusion model that combined the processing of multiple input modalities

Reliable Intelligence Identification on Vietnamese SNSs, VLSP 2020
Achieved a competitive performance (ranked 4th) in the VLSP shared task

Credit Scoring Challenge, Kalapa. Jsc 2020
Team ranked 1/146 with 800 contestants in constructing a banking credit scoring system

University of Liverpool - Ion Switching, Kaggle 2020
Bronze medalist (solo) in the University of Liverpool's Institute of Ageing and Chronic Disease competition

Practical deep learning, Vietnam Institute for Advanced Study in Mathematics 2019
Developed a deep learning-based Vietnamese spell corrector and ranked 1/50 in the final-project defense

National/international prizes

National Olympiad for Students 2017 - 2018
Vietnam Mathematical Society (VMS) Olympiad for universities and high school students
- Gold medal (2018), Silver medal (2017)

Vietnam Mathematics Olympiad (VMO) 2017 - 2018
The annual national mathematics competition for high school students
- Second prize (2018), Third prize (2017).
- International Mathematics Olympiad team selection test contestant

International Tournament of Towns (ITOT) 2017
International mathematical Olympiad for school students
- Bronze medal (2017).

Vietnam Institute for Advanced Study in Mathematics (VIASM) Scholarship 2016 - 2018
Scholarship of the National Program for the Development of Mathematics

Vallet Scholarship 2017 - 2018
Odon Vallet's scholarships for outstanding Vietnamese students

Professional services

- Reviewer for ICML, NeurIPS, ICLR, TPAMI, ICCV, ECCV, CVPR, IJCV, WACV, ACL, AAAI, KDD

Technical skills

- **Programming Languages**: Python, C, C++, Java
- **Frameworks**: TensorFlow, PyTorch, PyTorch Lightning, JAX, NumPy, etc